

DOEUN BUNHENG

Data Science

mrbunheng80@gmail.com | Phnom Penh, Kandal | +855-885-051-854 | +855-763-062-769
github.com/docunbunheng



SUMMARY

Fourth-year Data Science student with academic knowledge and project practice in Python, SQL, data analysis, machine learning, databases, and data visualization. Learned through university projects involving data cleaning, prediction models, natural language processing, classification, forecasting, and dashboard development. Eager to apply data-driven thinking to real business problems, improve practical analytical skills, and contribute to a professional team.

EDUCATION

Engineering Degree in Data Science

2022 – Expected 07/2027

Institute of Technology of Cambodia (ITC)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Database Systems, Statistical Modeling, Data Visualization, Probabilistic Graphical Models, Parallel & Distributed Systems

SKILLS

Data Analysis Tools	Python, SQL, R, Pandas, NumPy
Data Visualization	Matplotlib, Seaborn, Power BI, Plotly Dash
Machine Learning	scikit-learn, XGBoost, LSTM, GRU, CNN, BiLSTM, AlexNet
Databases	MySQL, PostgreSQL, Oracle 21c, MongoDB
Programming Languages	Python, SQL, R, C, C++
Frameworks	Django REST Framework, React, Node.js/Express
Big Data & Streaming	Apache Kafka (KRaft), PySpark Structured Streaming
Version Control	Git, GitHub
Soft Skills	Analytical Thinking, Problem Solving, Fast Learner, Team Collaboration

PROJECT EXPERIENCE

Lung Cancer Analysis Dashboard

05/2026 – 06/2026

Power BI, DAX, Python, Data Visualization

- Built a 6-page interactive Power BI dashboard analyzing 220,632 patient records across 25 countries and 24 variables.
- Designed structured pages: Overview, Smoking & Risk Factors, Treatment & Survival, Risk & Correlation Analysis, Healthcare & Environment, and Executive Summary.
- Created 12+ DAX measures including Diagnosis Rate %, Early Detection Rate by Healthcare Access, Smoker Diagnosis Rate, and Avg Survival Years.
- Identified smoking as the dominant risk factor (7.07% vs 2.05% for non-smokers — a 3.4× gap); all environmental factors clustered near the 4.06% population average.
- Found healthcare access drives a 3× early detection gap (60% good access vs 20% poor access), directly influencing survival outcomes.
- Built disconnected DAX tables for risk scenario modeling and heatmap-style matrix visuals with conditional formatting for cross-factor analysis.
- Delivered actionable public health insights aligned with UN SDGs 3, 10, and 17.

Men's Clothing Store Management System

04/2026 – 06/2026

Oracle 21c, PL/SQL, Django REST Framework, React, JWT

- Built a full-stack e-commerce platform with 3-tier architecture (Oracle database, Django REST API, React frontend) for a Cambodian men's retail store.
- Designed a normalized relational schema (3NF) with 7 tables, 4 views, and referential integrity through primary and foreign keys.
- Developed 6 PL/SQL triggers including 2 compound triggers to resolve the ORA-04091 mutating-table error for stock synchronization and order total calculation.
- Implemented 2 stored procedures with atomic transactions for customer registration and duplicate validation.
- Designed role-based access control with 3 Oracle roles (Admin, Sale, Customer) enforced at both database and API layers (defense in depth).
- Built REST API with Django REST Framework, python-oracledb driver, JWT authentication, and custom permission classes.
- Developed responsive React frontend; served as frontend lead on the `bunheng` branch using Git feature-branch workflow with code reviews via pull requests on GitHub.

Cambodia's Tourism Growth with Real-Time Weather Data

03/2026 – 05/2026

Apache Kafka, PySpark, Python, scikit-learn, Plotly Dash, Open-Meteo API

- Built an end-to-end AI-driven data pipeline streaming live weather data from Open-Meteo API into Apache Kafka, processed with PySpark

Structured Streaming across Bronze, Silver, and Gold Medallion layers.

- Designed a Star Schema data warehouse with 1 fact table (168 rows, 2012–2025) and 3 dimension tables derived from World Bank CSV and official MOT Cambodia PDF reports.
- Implemented Event Replay pattern with Kafka producers to simulate real-time airport immigration streaming at 1 record/second.
- Trained a RandomForest model achieving $R^2 = 0.89$ and MAE = 15,755 arrivals (2.9% error); lag-12 identified as the dominant feature at 80% importance.
- Generated 2026 tourist arrival forecasts for Cambodia’s three major cities using engineered lag features, YoY growth ratios, and seasonal indicators.
- Built an interactive Plotly Dash dashboard with 4 tabs: live weather charts (updated every 30 s), historical analytics, correlation scatter plots, and ML prediction sliders.
- Validated data quality with 5 automated checks before pipeline ingestion; collaborated in a 7-person team using Git feature-branch workflow.

Hourly Traffic Prediction at Urban Junction

Python, SARIMA, XGBoost, Random Forest, LSTM, GRU, Prophet — Joint Science Day

2024

- Analyzed 14,592+ hourly vehicle records; applied full preprocessing: cleaning, scaling, feature extraction, stationarity testing.
- Implemented and compared SARIMA, XGBoost, Random Forest, LSTM, GRU, and Prophet models.
- Designed deep learning sequence models with 24-hour input windows; evaluated using MAE, MSE, RMSE, R^2 , and MAPE.
- Achieved strong performance: GRU ($R^2 = 0.9579$), Random Forest (RMSE = 4.79); concluded deep learning and tree-based models outperform classical SARIMA.

CERTIFICATES

Scientifi-day

ASEAN Data Science Explorers

Course Python

Techpreneur Bootcamp

LANGUAGES

Khmer: Native

English: Working Proficiency (Listening: Good, Speaking: Very Good, Writing: Good)

French: Low

STRENGTHS

Analytical Thinking

Problem Solving
Fast Learner

Research-Oriented Mindset
Team Collaboration